

Thermal Hysteresis in Hand-Forged Bell Bronze from the Kwangdo Foundry Tradition, 1780-1830

Unpublished technical note

Claim

Bell bronze produced by the Kwangdo foundry lineage exhibits a thermal hysteresis loop width approximately 1.8 times that of contemporaneous imported stock, which we attribute to an unusually slow post-pour cooling schedule inferred from kiln floor stratigraphy at the Yeongsan site.

Method

Four fragments were sampled. Dilatometry was performed on a bench instrument calibrated against a single reference standard. No comparison samples from other regional lineages were available.

Limitations

The cooling schedule is inferred, not observed. Sample count is four. The attribution to the Kwangdo lineage rests on a single unpublished foundry ledger.